

Data representation

1> <https://www.techtarget.com/searchstorage/definition/byte>

2> <https://web.stanford.edu/class/cs101/bits-bytes.html>

Video: <https://www.youtube.com/watch?v=KLR5Y2uBYng>

Bit, Byte, Convert Base 2 to Base 10 (Decimal)

- Computers are electronic devices that store voltages. These devices are either ON or OFF. Thus only 2 voltages are possible, 1 for ON and 0 for OFF. Thus, computer stores information (data) as two symbols ie 0 and 1.
- Data (0 and 1) is a physical concept, a raw sequence of bits and bytes stored on a computer. Data has no inherent meaning and must be interpreted by whoever is reading it. Computers store data using a sequence of 1's and 0's. Later, they read back the 1's and 0's and try to recreate the abstract concept from the raw data. Depending on the assumptions made, the 1's and 0's can mean very different things.
- What is a Bit? (The Single Switch)**
 - A **bit** (short for **binary digit**) is the smallest piece of information a computer can understand.
 - A computer's brain is made of billions of microscopic electronic switches. Because a switch can only be in two positions, a bit can only have one of two values: **0** = Off (Low voltage / False) **1** = On (High voltage / True)
 - A single bit cannot tell you much. It can only answer simple "yes or no" questions.
- What is a Byte? (The Character Builder)**
 - Because one bit is too small to be useful on its own, computers bundle them into groups of eight. With 8 bits combined, you can create **256 different combinations** of zeros and ones (from 00000000 to 11111111)
 - A **byte** is a group of **8 bits** working together. Here is how a single byte is structured and represented. A byte consists of 8 slots. Each slot is a bit that holds a value of either 0 or 1.

	1 Byte							
	Upper Nibble				Lower Nibble			
Bit position	Bit -7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Binary number	MSB							LSB
	1	0	0	1	0	1	1	0
Base-2 Place value	1×2^7 = 128	1×2^6 = 64	1×2^5 = 32	1×2^4 = 16	1×2^3 = 8	1×2^2 = 4	1×2^1 = 2	1×2^0 = 1
Convert Base-2 to Base-10(decimal)	1×2^7 = 128	0×2^6 = 0	0×2^5 = 0	1×2^4 = 16	0×2^3 = 0	1×2^2 = 4	1×2^1 = 2	0×2^0 = 0
Value(decimal)	150							

- Computers use a Base-2 (Binary) place value system, Meaning each slot's value doubles from right to left, starting at 1.

Bit Position	8	7	6	5	4	3	2	1
Place Value	128	64	32	16	8	4	2	1
Bit State	0	1	0	0	0	0	0	1

- Bits are numbered from right-to-left. Bit 0 is the rightmost and the smallest; bit 7 is leftmost and largest. The "leftmost" bit in a byte is the biggest.
- Bit numbering is bit positions in a binary number. The bit number starts with zero and is incremented by one for each subsequent bit position. The value of the number is calculated by using exponentiation with the bit number and a base of 2.

- These bits are grouped as

- Nibble (4 bits): A nibble is a sequence of 4 bits

- 8 bits makes 1-byte

8-bits = 1-byte = 1101 0010

- 16 bits makes 1-word

16bits = 1 word 0000 0010 1101 0010

- 32 bits makes 1-double word

32bits = 1 double word 0000 0000 0000 0010 1101 0010

- Example

The binary sequence 00001001 is decimal number 9.

	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
binary sequence	0	0	0	0	1	0	0	1
multiply by place holder.	0×2^7	0×2^6	0×2^5	0×2^4	1×2^3	0×2^2	0×2^1	1×2^0
	0	0	0	0	8	0	0	1
Decimal number	9							

- Range of 1 byte variable.

A byte can represent any whole number from **0 to 255**.

00000000 = 0

01000001 = 65

11111111 = 255 (All switches on: $128+64+32+16+8+4+2+1$)

Text Representation (ASCII Code)

Computers use a translation chart called **ASCII** (American Standard Code for Information Interchange) to map numbers to keyboard characters.

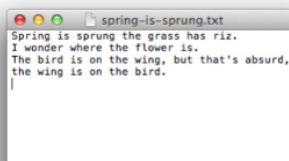
- The byte 01000001 equals 65.
- On the ASCII chart, 65 is the capital letter "A".
- The word "HI" requires two bytes: 01001000 (H) and 01001001 (I)
- One byte works well for individual characters.

ASCII TABLE

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	\$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[END OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

Typing, Bytes, and You

- Each letter is stored in a byte, as below
- 100 typed letters takes up 100 bytes
- When you send, say, a text message, the numbers are sent
- Text is quite compact, using few bytes, compared to images etc.



Underlying bytes in RAM

S	p	r	i	...
83	112	114	105	

Visualizing the Scale

To see how this grows into the files you use every day, look at how bytes multiply:

- **1 Bit:** A single light switch (On or Off).
- **1 Byte (8 bits):** A single letter (e.g., "H").
- **1 Kilobyte (KB, ~1,000 bytes):** A short paragraph of text.
- **1 Megabyte (MB, ~1 million bytes):** A 3-minute MP3 song or a high-quality photo.
- **1 Gigabyte (GB, ~1 billion bytes):** A 1-hour streaming movie